

The Text on the Creative

An under-exploited ranking modality for short-form video ads.

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Meta, Instagram Ads Ranking. Underlined names are presenting.



Poster, paper and summary
shubham.gg/recsys26

A video ad tells you what it's selling, in text, right on the frame. The ranker was **blind to it**. We made it a feature, at 3 bytes per ad.

0.06–0.08%

Online NE gain, multi-week 80%-traffic A/B, $p < 0.01$

$\approx 0.12\%$

Offline NE gain on Reels-class video, where overlay text is densest

$\sim 500\times$

Less storage per ad: 3-byte codes vs a 1,536-byte dense embedding

NE = normalized entropy: log loss divided by the entropy of the average CTR. Lower is better; gains are relative NE reductions vs the production model.

How it works

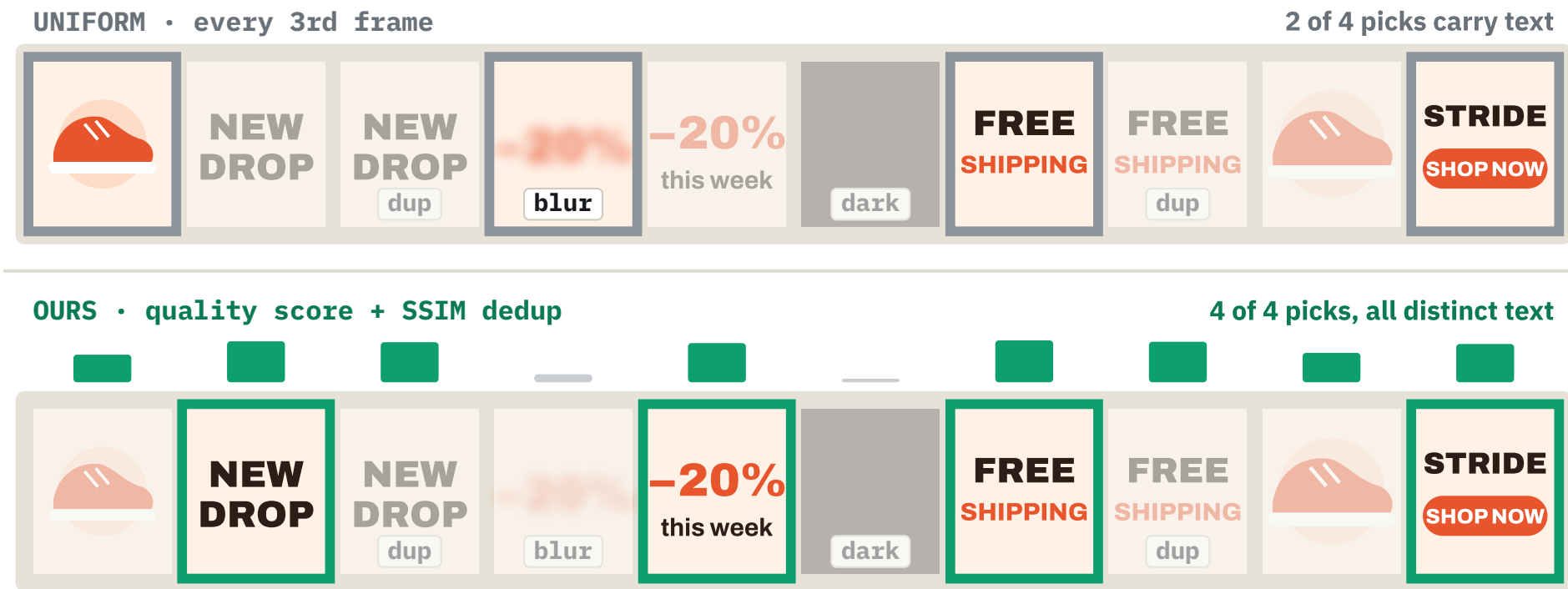
Should also work for ASR transcripts, scene tags, landing-page text and logos.



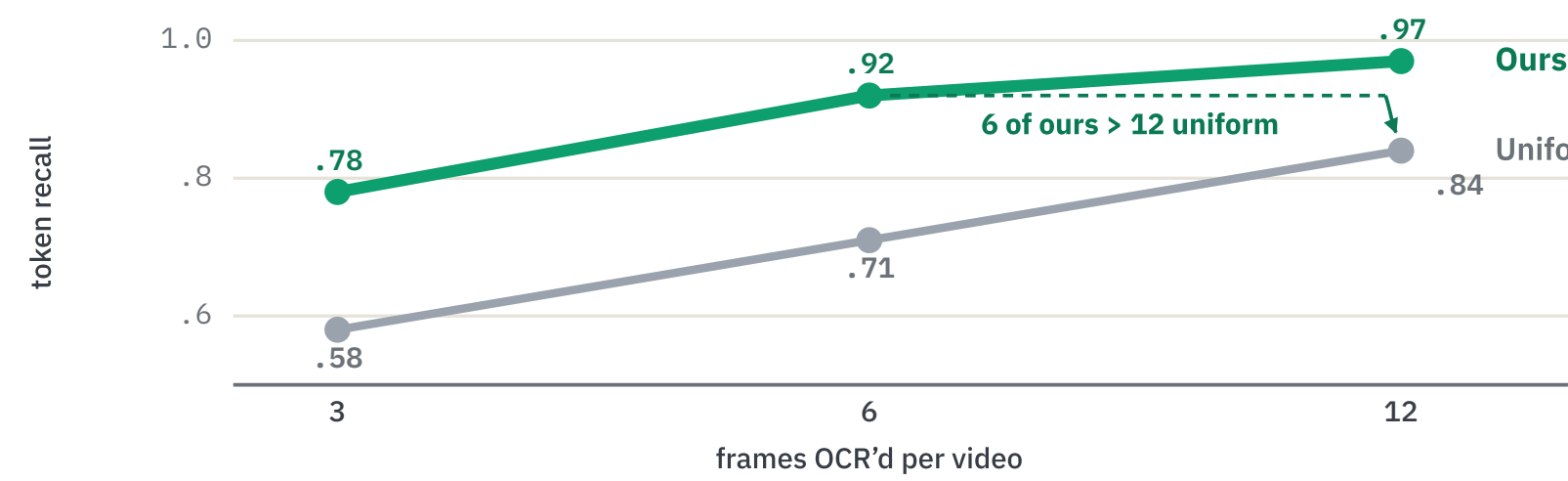
01 FRAME SELECTION

OCR fewer, better frames.

We reuse the **CTR-trained quality scorer** from thumbnail selection to rank frames, so there's **no extra inference cost**. Whole-frame similarity (SSIM) drops near-duplicates, and we sample candidates densely early in the video, sparsely later.



Illustrative: one ad, same 4-frame budget. Paper Fig. 2.

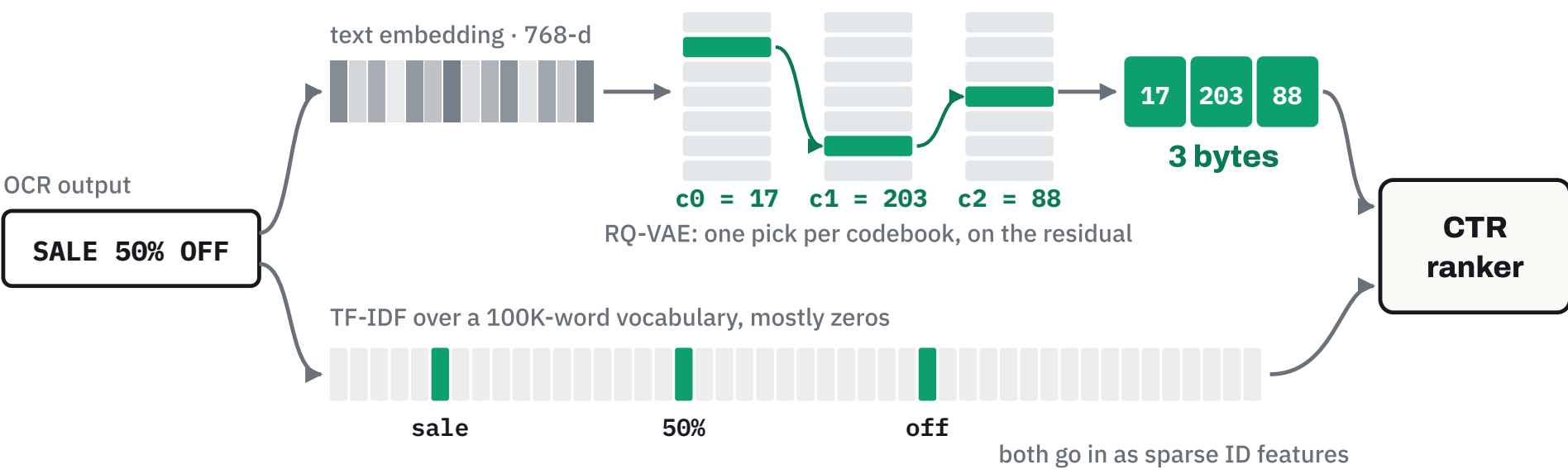


OCR token recall vs an every-1s oracle, 5,000 held-out videos. **Half the frames, higher recall**. Paper Table 4.

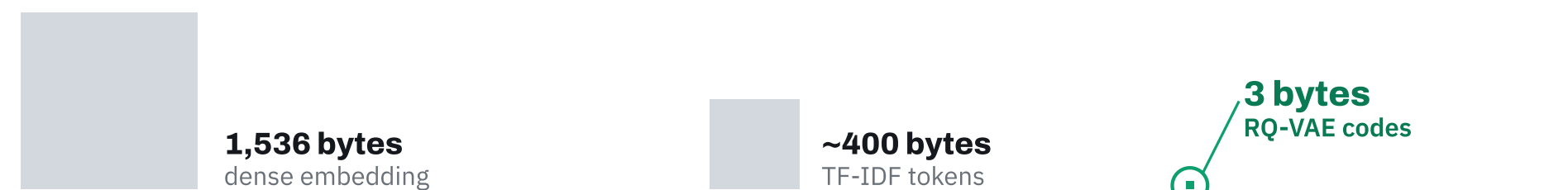
03 REPRESENTATION

Compress the text into 3 bytes of ID features.

A residual-quantized VAE (**RQ-VAE**) turns the ad's text embedding into three 1-byte codes (semantic IDs); **TF-IDF** keeps the exact words the codes lose. Both go into the ranker as sparse ID features. On top of both, the dense embedding added no lift beyond noise, so we dropped it.



Two views of the same OCR text: 3 codes and a sparse TF-IDF vector.



Square area $\propto$ bytes stored per ad. **Codes are $\sim 500\times$ smaller than the dense embedding**. Paper Table 10.

HOW WE COMPARED THEM LLM-judged neighbor coherence and linear probes over **200K ads**, lexical and reconstruction checks with no judge, and **5 human reviewers** (Spearman $\rho = 0.74$ vs the judge).

Does your ranker read this?

Ours does now.

Our in-house Ads-OCR reads the on-screen text. The features are live in Instagram ads ranking.



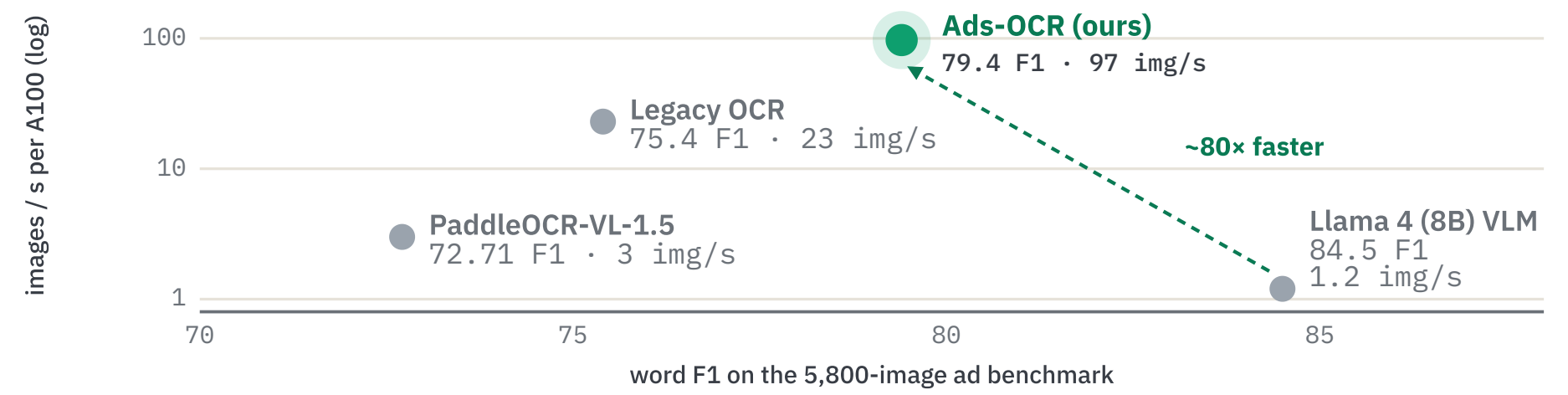
02 PERCEPTION

An in-house OCR model, not a general VLM.

Ads-OCR is built for ad creatives and covers **41 languages**. Detection, cropping and recognition run word-parallel in one batched pass, with FlashAttention, FP16 and torch.compile.



What each stage of Ads-OCR does to one frame, in a single batched pass. Paper Fig. 1.

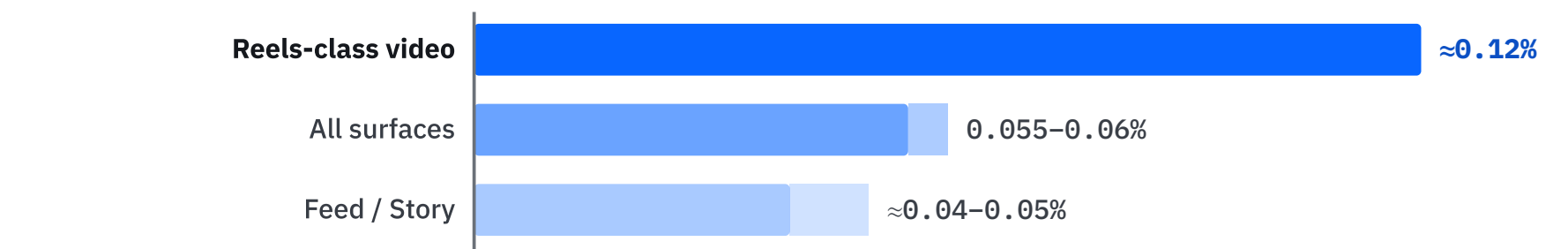


93% of the VLM's word F1 at $\sim 80\times$ the throughput. Paper Table 1.

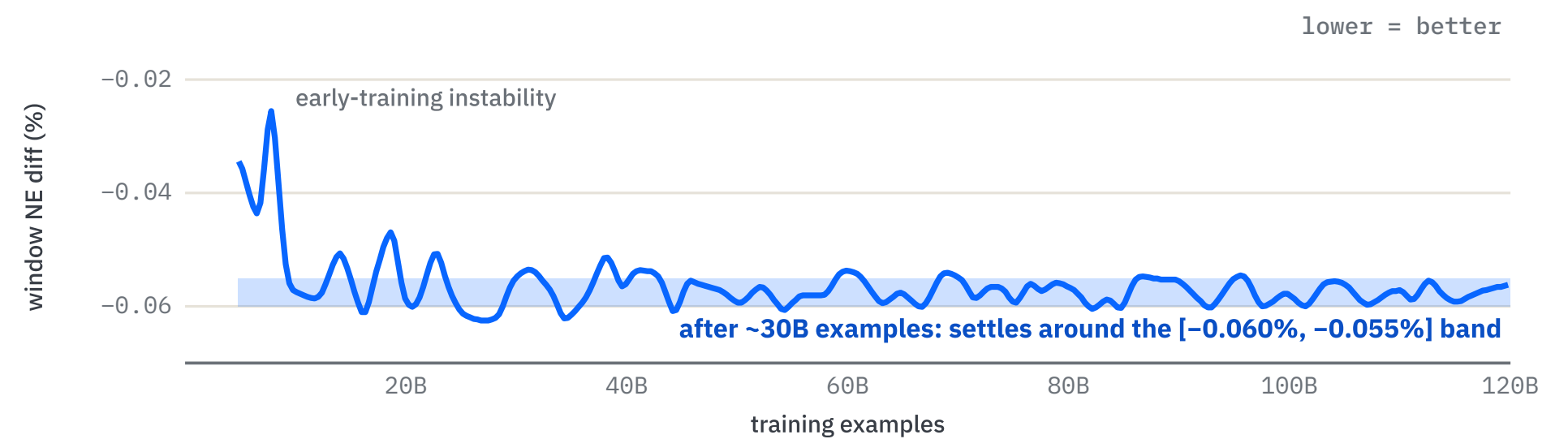
04 RESULTS

Offline gains held up online.

Offline, the codes alone give **0.045–0.05%** and TF-IDF alone **0.015–0.02%**; together **0.055–0.06%**, and **$\approx 0.12\%$** on Reels-class video. Online, a multi-week A/B on 80% of traffic gave **0.06–0.08%**, $p < 0.01$.



Offline NE gain by surface. The pale end of a bar is the reported range. Paper §8.1.



Window NE diff vs production across $\sim 120B$ training examples (diff = -gain). Paper Fig. 3.